

A Commodity-Augmented Dynamic Factor Model for Australian GDP

1 Introduction

This paper adapts the Stock–Watson (1991) coincident-indicator framework to nowcast Australia’s real GDP growth, augmenting the baseline specification with commodity and trade indicators to better capture the structural drivers of the Australian business cycle. Australia makes for a compelling out-of-sample test of the framework: as a resource-exporting, financially open economy deeply integrated into Asian commodity markets, its cycle is shaped as much by Chinese steel demand, iron ore pricing and terms-of-trade swings as by domestic conditions, making a purely conventional indicator set an impoverished description of its macroeconomic dynamics. By incorporating commodity export data alongside the standard activity proxies, we aim to provide a richer, more country-appropriate signal for the latent common factor that reflects both the domestic and external engines of Australian growth. The model summarizes the co-movement of one quarterly target series and seven monthly indicators (Table 1) into a single latent common factor f_t , estimated recursively via the Kalman filter and obtained by maximum likelihood.

Table 1: Data Description

Variable	Unit	Transformation	Type	Native sample
GDP	Chained volume, 2019=100	$\Delta \log$	Hard (target)	Q3-1960 – Q1-2026
Industrial Production (IP)	Index, 2010=100	$\Delta \log$	Hard	08-1974 – 02-2026
Construction (dwellings approved)	Number, SA	$\Delta \log$	Hard	07-1986 – 04-2026
New motor vehicle sales	Units	$\Delta \log$	Hard	01-1994 – 05-2026
Employment	Thousand	$\Delta \log$	Hard	02-1978 – 02-2026
Stock index (ASX)	Index, USD	$\Delta \log$	Hard	06-2000 – 05-2026
Total Exports: Commodities	AUD billion	$\Delta \log$	Hard	01-1955 – 03-2026
Composite Leading Indicator (CLI)	Index	log	Leading (t+3)	02-1966 – 04-2026

¹ ADF tests indicate that all variables are I(1) except the Composite Leading Indicator, which is I(0) by construction and is accordingly entered in log-levels rather than log-differences.

² *Sources:* GDP, Industrial Production, Employment, Construction and Vehicle sales from the Australian Bureau of Statistics abs2026; Total Commodity Exports from the ABS Balance of Payments release abs2026bop; Stock index (ASX) from Bloomberg bloomberg2026; Composite Leading Indicator from the OECD Main Economic Indicators oecd2026.

Although several indicators are individually available from the 1950s–1970s, the stock index (ASX) is the binding constraint, as it starts in June 2000. We include the ASX because its original daily frequency provides higher-frequency information that may help capture more timely changes in economic activity. To incorporate it consistently into the monthly dynamic factor model, the daily series was converted into a monthly series by taking end-of-month closing values. The model is therefore estimated on the monthly sample from **June 2000 to May 2026**, with quarterly GDP and missing observations handled by the Kalman filter.

2 Indicator Selection

Indicator selection was driven by the research question: which series carry genuine information about the commodity and trade channels of the Australian cycle, over and above what standard domestic proxies already provide? Industrial production and employment form the domestic activity backbone, retained for comparability with the baseline specification and as a check that augmentation does not come at the cost of the standard cyclical signal. Dwelling approvals add a housing dimension specific to Australia, where residential construction is among the most cyclically sensitive GDP components and approvals lead actual building activity by several months. New motor vehicle sales proxy private consumption of durables, one of the first expenditure categories to respond as household sentiment shifts, providing a timely read on domestic demand conditions. The commodity augmentation enters through total commodity exports, which directly capture the external revenue channel that distinguishes Australia’s cycle from that of a closed or manufacturing-led economy. Since iron ore, coal and LNG collectively account for a disproportionate share of export earnings, movements in this series reflect both global commodity demand from Asian markets and Australia’s terms of trade simultaneously, making it a compact summary of the external impulse. The ASX stock index adds a forward-looking financial signal that may anticipate turning points ahead of the real-side indicators. Finally, the OECD Composite Leading Indicator is included not as a contemporaneous proxy but as a genuine leading variable, entering the measurement equation

on f_{t+3} rather than f_t , providing a forward horizon that the monthly activity indicators cannot supply on their own.

2.1 Validation: Principal Component Analysis

To assess whether the proposed indicator set actually spans the commodity and trade dimensions of the Australian cycle, we use PCA as a pre-estimation diagnostic. Under the assumption that each series is a noisy partial signal of a latent business-cycle state, the first principal component identifies the linear combination that explains the largest share of joint variation, and the resulting loadings reveal which series track the common cycle versus moving on idiosyncratic dynamics.

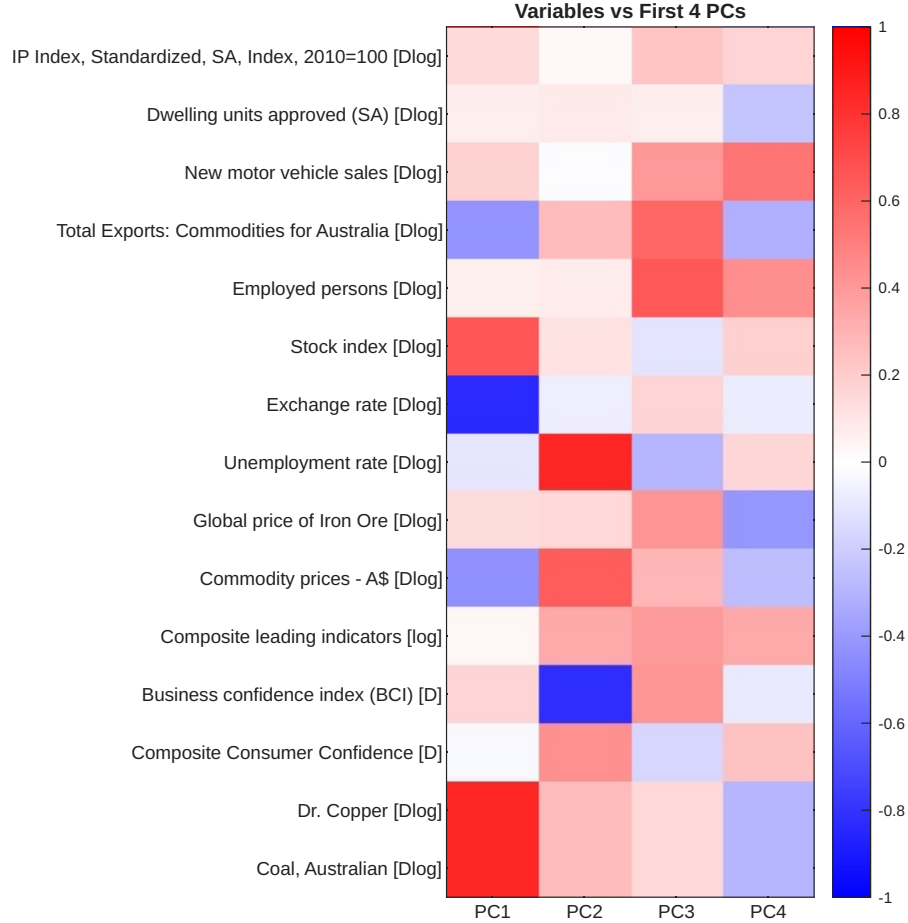


Figure 1: PCA Correlation Heatmap

The dominant loadings on PC1 fall on coal prices, the exchange rate and the ASX stock index, consistent with a cycle driven primarily by commodity prices and external financial conditions rather than domestic production alone. This supports the augmentation strategy: the first common factor in Australian data is not an industrial production factor but a terms-of-trade and financial conditions factor, and an indicator set that omits these dimensions would systematically mischaracterize the cycle. The PCA also informs exclusion decisions. Copper loads strongly on PC1, but through a global rather than Australia-specific channel – Australia is not a significant copper producer or trader, and its loading likely proxies the global commodity cycle rather than any domestic transmission mechanism. Including it would risk conflating global and Australian signals in the factor. Coal and iron ore load strongly for the right reasons, but their information content is already embodied in the aggregate commodity export series and the exchange rate, making separate foreign benchmark prices redundant. We therefore treat strong loadings on commodity and trade-related series as confirmation that our chosen indicators span the relevant dimensions, while using the same evidence to exclude duplicates and globally-priced benchmarks without a direct Australian transmission channel.

3 Model Specification

Each (de-measured, differenced) series $y_{i,t}$ is assumed to load on a single common factor and an idiosyncratic AR(2) component,

$$y_{i,t} = \lambda_i f_t + e_{i,t}, \quad f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + w_t, \quad e_{i,t} = \psi_{i,1} e_{i,t-1} + \psi_{i,2} e_{i,t-2} + \varepsilon_{i,t},$$

written in state-space form as $\mathbf{y}_t = \mathbf{H}\mathbf{h}_t + \mathbf{w}_t$, $\mathbf{h}_t = \mathbf{F}\mathbf{h}_{t-1} + \mathbf{v}_t$, $\mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$, with $\mathbf{h}_t$ the 27×1 state vector and $\mathbf{y}_t$ the 8×1 observation vector:

$$\mathbf{y}_t = \begin{pmatrix} \Delta \log(\text{GDP}_t) \\ \Delta \log(\text{IP}_t) \\ \Delta \log(\text{Con}_t) \\ \Delta \log(\text{Veh}_t) \\ \Delta \log(\text{Emp}_t) \\ \Delta \log(\text{Stk}_t) \\ \Delta \log(\text{Exp}_t) \\ \log(\text{CLI}_t) \end{pmatrix}, \quad \mathbf{h}_t = \begin{pmatrix} f_{t+3}, f_{t+2}, f_{t+1}, f_t, f_{t-1}, f_{t-2}, f_{t-3}, f_{t-4}, \\ e_{\text{GDP},t}, e_{\text{GDP},t-1}, e_{\text{GDP},t-2}, e_{\text{GDP},t-3}, e_{\text{GDP},t-4}, \\ e_{\text{IP},t}, e_{\text{IP},t-1}, e_{\text{Con},t}, e_{\text{Con},t-1}, e_{\text{Veh},t}, e_{\text{Veh},t-1}, \\ e_{\text{Emp},t}, e_{\text{Emp},t-1}, e_{\text{Stk},t}, e_{\text{Stk},t-1}, e_{\text{Exp},t}, e_{\text{Exp},t-1}, \\ e_{\text{CLI},t}, e_{\text{CLI},t-1} \end{pmatrix}_{27 \times 1}^\top.$$

The factor block is extended three steps into the future ($f_{t+3}, f_{t+2}, f_{t+1}$) purely so that the CLI can load on f_{t+3} as a named coordinate of today's state, while GDP – being a quarterly flow – continues to load on a five-month, triangularly-weighted window of monthly factor values ($\frac{1}{3}, \frac{2}{3}, 1, \frac{2}{3}, \frac{1}{3}$ on $f_t, \dots, f_{t-4}$), with the same weights applied to its own idiosyncratic component, following the same temporal-aggregation device used in the baseline code, in the spirit of Camacho and Pérez-Quiros's (2010) Euro-STING methodology. Every other indicator loads contemporaneously on f_t alone and carries a simple two-lag idiosyncratic AR(2). The explicit measurement and transition equations implementing this structure are given in Appendix A.

$$\mathbf{Z} = (\lambda_{\text{GDP}}, \lambda_{\text{IP}}, \lambda_{\text{Con}}, \lambda_{\text{Veh}}, \lambda_{\text{Emp}}, \lambda_{\text{Stk}}, \lambda_{\text{Exp}}, \lambda_{\text{CLI}}, \phi_1, \phi_2, \{\psi_{i,1}, \psi_{i,2}\}_{i \in \{\text{GDP}, \text{IP}, \text{Con}, \text{Veh}, \text{Emp}, \text{Stk}, \text{Exp}, \text{CLI}\}}, \{\sigma_{\varepsilon,i}\}_{i \in \{\text{GDP}, \dots, \text{CLI}\}})^\top$$

The full vector of freely estimated parameters has 34 parameters in total. Identification of the factor's scale is achieved by *fixing* the variance of the factor's own innovation at unity ($\sigma_w^2 \equiv 1$, not estimated) rather than by normalizing any loading – λ_{GDP} is therefore freely estimated like every other loading, and is reported among the results below. The complete table of 34 point estimates and standard errors is given in Appendix C.

4 Results

Maximum likelihood estimation converges successfully, with `fminunc` recovering a well-identified parameter vector and finite-difference Hessian standard errors that are precise throughout. As an internal consistency check, the correlation between the full fitted GDP series (factor *and* idiosyncratic component) and observed GDP growth is 1 at all quarters where GDP is observed, confirming the filter reconstructs the data it was given. More substantively, the correlation between observed GDP growth and the *factor-only* fitted series is 0.9617, indicating that the common cycle alone accounts for the bulk of co-movement in Australian GDP growth and that the augmented indicator set is extracting a genuinely informative signal.

Table 2 reports factor loadings and associated significance tests. Employment and GDP carry the largest and most precisely estimated loadings, significant well below the 1% level, confirming that the labor market remains the single strongest contemporaneous signal of the Australian cycle. Commodity exports load positively and significantly at the 5% level ($\lambda_{\text{Exp}} = 0.1396$, $p = 0.025$), providing direct evidence that the trade and commodity channel carries incremental information about the common factor beyond what domestic indicators alone would supply, the central empirical contention of this paper. Vehicle sales and construction are also significant, consistent with the view that household consumption of durables and residential investment are meaningfully cyclical in Australia.

Table 2: Factor loading significance tests.

Variable	Factor loading	Std. error	t-stat	p-value
GDP	0.5995	0.0457	13.125	<0.001
IP	0.0081	0.0439	0.184	0.854
Construction	0.1469	0.0442	3.324	0.001
Vehicle sales	0.3639	0.0458	7.937	<0.001
Employment	0.8531	0.0652	13.084	<0.001
Stock index	0.0155	0.0698	0.222	0.824
Exports	0.1396	0.0624	2.238	0.025
CLI	0.0087	0.0091	0.957	0.339

Three indicators, namely industrial production, the ASX stock index and the CLI, load weakly and insignificantly on the common factor. This is a substantively interesting finding rather than simply a diagnostic failure. For industrial production, the result likely reflects Australia’s relatively small manufacturing base: unlike the United States, where IP is a core coincident indicator, Australian industrial output is narrow and volatile enough that its idiosyncratic variation dominates any common-cycle signal. The stock index’s insignificance is perhaps more surprising given its strong PC1 loading in the pre-estimation PCA, but may reflect the well-documented disconnect between equity valuations and real activity over this particular sample, which spans two major asset price cycles. The CLI’s near-zero loading suggests its forward-looking content is not well captured by the contemporaneous factor structure, even with the three-month lead built into the measurement equation, which motivates further investigation into alternative leading indicator specifications.

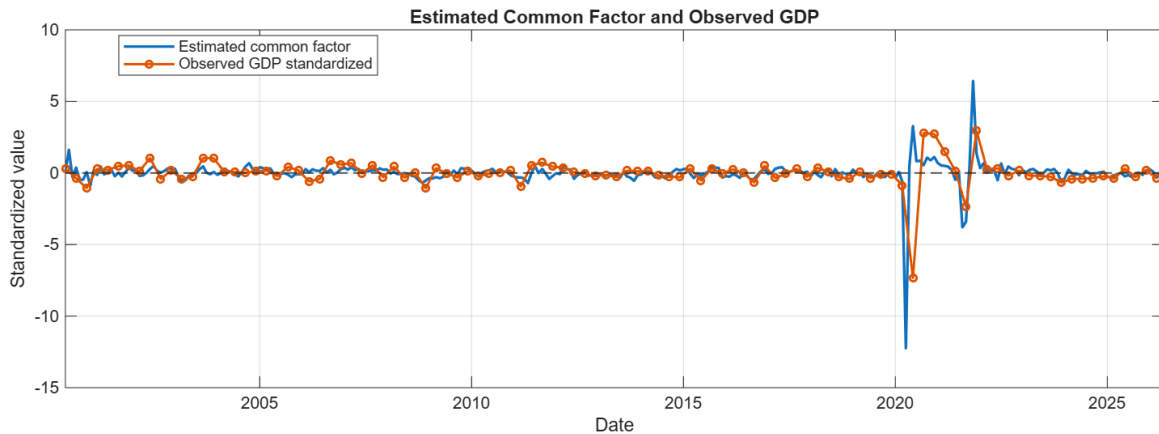


Figure 2: Estimated common factor versus standardised observed GDP growth.

Figure 2 plots the estimated latent factor against standardized GDP growth. The two series co-move closely around the major turning points in the sample, most visibly the 2020 pandemic contraction and the subsequent recovery, which is reassuring: the factor is picking up genuine, broad-based cyclical movement rather than being driven by any single series. Some divergence persists in calmer periods, consistent with the presence of GDP-specific idiosyncratic variation that the common factor is not designed to capture.

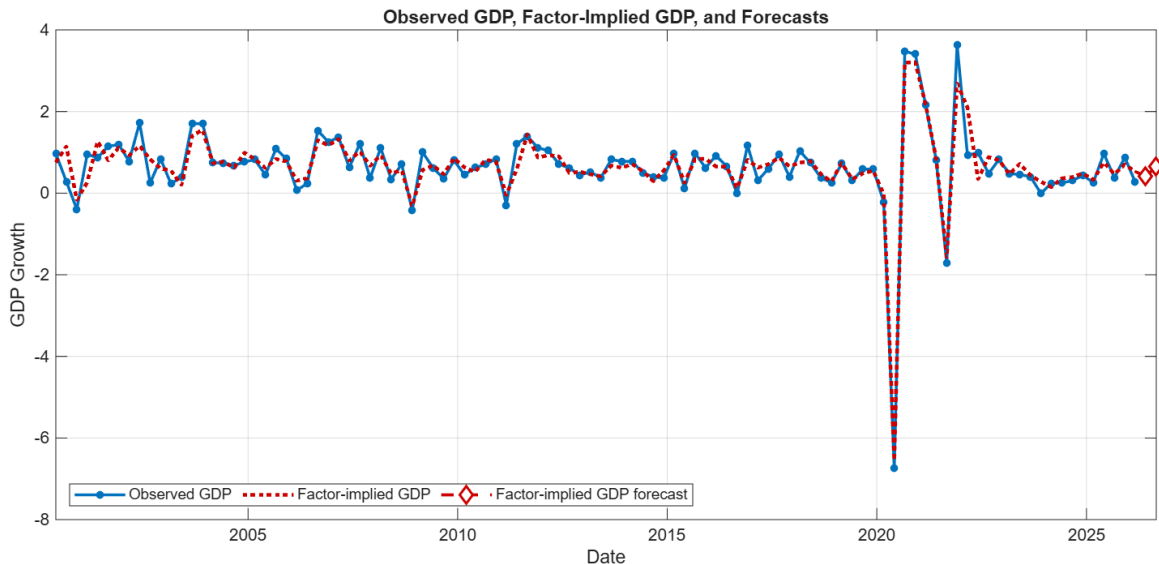


Figure 3: Observed GDP growth, factor-implied GDP growth and factor-implied forecasts.

Figure 3 compares the factor-only fitted series against observed quarterly GDP growth and extends the horizon to the next two quarterly releases. The fit is tightest around recessions and sharp cyclical turning points, reinforcing the interpretation that the common factor captures the signal most reliably precisely when it matters most for real-time monitoring. The remaining gaps in expansion periods likely reflect GDP-specific noise not shared with the broader indicator set. The factor-only state forecast implies Australian real GDP growth of **0.41%** for Q2-2026 and **0.65%** for Q3-2026, pointing to a modest but stable recovery in activity consistent with a soft-landing scenario.

5 Conclusion

This paper has shown that augmenting the Stock–Watson (1991) coincident-indicator framework with commodity and trade data produces a coherent and informative nowcasting model for the Australian economy. The key finding is that commodity exports load significantly on the common factor, providing direct evidence that the external, terms-of-trade-driven channel carries genuine cyclical information beyond what a standard domestic indicator set would capture, a result with direct relevance for monitoring resource-export-dependent economies more broadly. The model’s performance is strongest around major cyclical turning points, which is precisely the environment in which accurate real-time monitoring has the highest value for policy and investment decisions. The insignificance of industrial production and the stock index over this sample points to meaningful differences in the transmission of business cycle dynamics between Australia and the US economy for which the baseline framework was originally designed, and suggests that country-specific indicator selection is not merely a refinement but a precondition for reliable nowcasting outside the original application.

Two natural extensions follow from these results. First, a formal pseudo-real-time forecasting evaluation against a domestic-only baseline would allow a direct test of whether commodity augmentation reduces nowcast error in a statistically meaningful sense. Second, the framework could be extended to other commodity-exporting economies such as Chile, Canada, Norway or the Gulf states, where similar terms-of-trade dynamics shape the business cycle, to assess whether the results generalize across different resource endowments and institutional settings.

B Figures

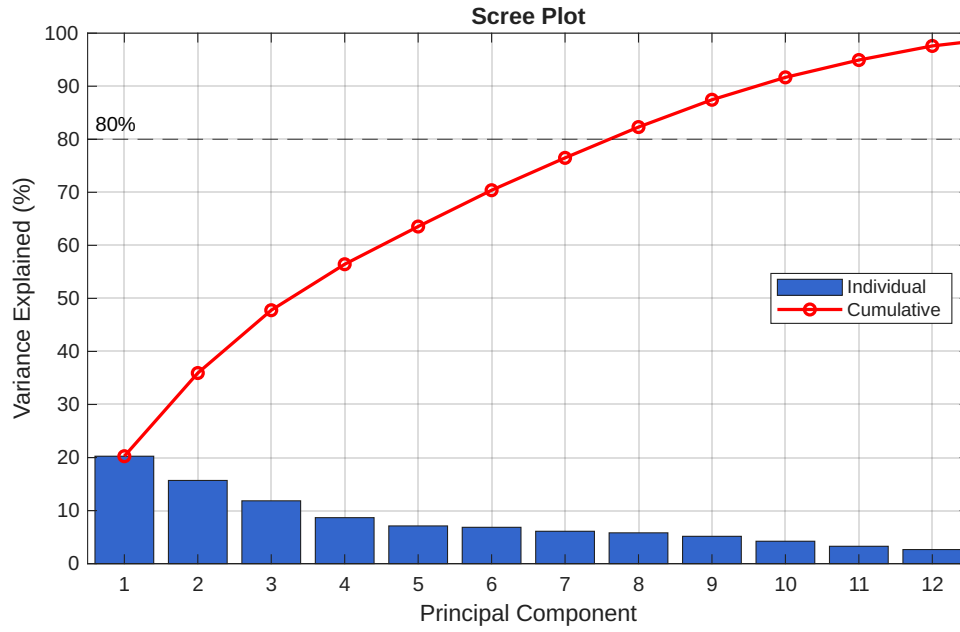


Figure 4: Principal Component Analysis Variance Explanation

C Full parameter estimates

Table 3: Maximum-likelihood estimates and Cramér–Rao standard errors for all 34 freely estimated parameters in $\mathbf{Z}$.

Param.	Estimate	Std. error	Param.	Estimate	Std. error
λ_{GDP}	0.5995	0.0457	$\psi_{\text{Veh},2}$	-0.4559	0.0544
λ_{IP}	0.0081	0.0439	$\psi_{\text{Emp},1}$	0.0548	0.1002
λ_{Con}	0.1469	0.0442	$\psi_{\text{Emp},2}$	0.3903	0.1357
λ_{Veh}	0.3639	0.0458	$\psi_{\text{Stk},1}$	0.1660	0.0617
λ_{Emp}	0.8531	0.0652	$\psi_{\text{Stk},2}$	-0.1046	0.0581
λ_{Stk}	0.0155	0.0698	$\psi_{\text{Exp},1}$	-0.0294	0.0589
λ_{Exp}	0.1396	0.0624	$\psi_{\text{Exp},2}$	-0.0849	0.0587
λ_{CLI}	0.0087	0.0091	$\psi_{\text{CLI},1}$	1.6694	0.0389
ϕ_1	0.1752	0.0904	$\psi_{\text{CLI},2}$	-0.7504	0.0388
ϕ_2	-0.2227	0.0853	σ_{GDP}	0.1590	0.0560
$\psi_{\text{GDP},1}$	0.6231	0.0935	σ_{IP}	0.6934	0.0288
$\psi_{\text{GDP},2}$	-0.8357	0.1013	σ_{Con}	0.8727	0.0364
$\psi_{\text{IP},1}$	0.8371	0.0600	σ_{Veh}	0.6494	0.0295
$\psi_{\text{IP},2}$	-0.1564	0.0605	σ_{Emp}	0.4910	0.0764
$\psi_{\text{Con},1}$	-0.5423	0.0574	σ_{Stk}	0.9791	0.0405
$\psi_{\text{Con},2}$	-0.2329	0.0575	σ_{Exp}	0.9979	0.0415
$\psi_{\text{Veh},1}$	-0.0533	0.0580	σ_{CLI}	0.1963	0.0082

Table 4: Correlation of Each Variable with the Estimated Common Factor (PC1)

Variable	Corr. with PC1
Coal, Australian [$\Delta \log$]	+0.845
Dr. Copper [$\Delta \log$]	+0.845
Exchange rate [$\Delta \log$]	−0.832
Stock index [$\Delta \log$]	+0.663
Commodity prices – A\$ [$\Delta \log$]	−0.434
Total Exports: Commodities for Australia [$\Delta \log$]	−0.414
New motor vehicle sales [$\Delta \log$]	+0.178
Business confidence index (BCI) [Δ]	+0.168
IP Index, Standardized, SA [$\Delta \log$]	+0.143
Global price of Iron Ore [$\Delta \log$]	+0.135
Unemployment rate [$\Delta \log$]	−0.095
Dwelling units approved (SA) [$\Delta \log$]	+0.063
Employed persons [$\Delta \log$]	+0.062
Composite leading indicators [$\log$]	+0.035
Composite Consumer Confidence [Δ]	−0.025

References

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